test-example

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test-example

High-performance Computing
Example Class Test

50 minutes

This test is worth 10% of the mark for this module.

Instructions

You should write a single C++ program to address the task below.

All code should be contained in a single `.cpp` file.

Your code **must** compile and run correctly on the provided Linux environment (spitfire/hurricane).

Your code (the .cpp file) should be submitted to Blackboard Learn by the end of the test.

Task

The vector $\mathbf{f}$ denotes a function $f(x)$, expressed at N_x grid-points x_i, with spacing h, on the interval $0 \le x \le L$.

The second-derivative operator on these grid-points can be approximated to second-order accuracy using the forward stencil:

$$f''(x) \approx g(x) = \frac{2f(x) - 5f(x+h) + 4f(x+2h) - f(x+3h)}{h^2}$$

Given $\mathbf{f}$, the discrete second derivative $\mathbf{g}$ can be evaluated using the matrix-vector equation

$$\mathbf{g} = \mathbf{D}\mathbf{f} + \mathbf{b} \qquad (1)$$

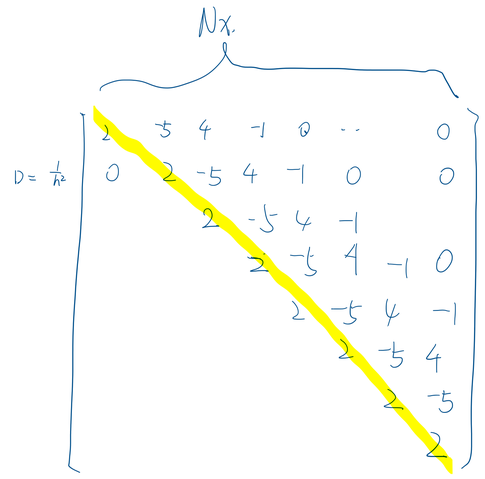
$$\mathbf{D} = \frac{1}{h^2}\begin{pmatrix} 2 & -5 & 4 & -1 & 0 & \dots & \dots & 0 \\ 0 & 2 & -5 & 4 & -1 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & 2 & -5 & 4 & -1 & 0 \\ \vdots & & & \ddots & 2 & -5 & 4 & -1 \\ \vdots & & & & \ddots & 2 & -5 & 4 \\ \vdots & & & & & \ddots & 2 & -5 \\ 0 & \dots & \dots & \dots & \dots & \dots & 0 & 2 \end{pmatrix} \quad \mathbf{b} = \frac{1}{h^2}\begin{pmatrix} 0 \\ \vdots \\ \vdots \\ \vdots \\ 0 \\ -f(L+h) \\ 4f(L+h) - f(L+2h) \\ -5f(L+h) + 4f(L+2h) - f(L+3h) \end{pmatrix} \} N_x$$

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Write a program which computes the second derivative of the function $f(x) = \sin(\pi x)$ on the interval $0 \le x \le 1$ using the approach above. Use $N_x = 1001$ grid points, such that $h = 0.001$. Your program should:

1. Generate the vector $\mathbf{f}$ by evaluating the function $f(x)$ on the grid points x_i. [10%]
2. Generate the matrix $\mathbf{D}$ in an efficient storage format and generate the RHS vector $\mathbf{b}$. [50%]
3. Use a BLAS routine appropriate to your chosen storage format to evaluate Equation (1). [20%]
4. Evaluate the exact solution $f''(x) = -\pi^2 \sin(\pi x)$ as $\mathbf{f}''$ and use BLAS to compute $\epsilon = ||\mathbf{f}'' - \mathbf{g}||_2$. [20%]

END OF TEST



$$b = \frac{1}{h^2}\begin{bmatrix} 0 \\ \vdots \\ 0 \\ -f(L+h) \\ 4f(L+h) - f(L+2h) \\ -5f(L+h) + 4f(L+2h) - f(L+3h) \end{bmatrix}$$

$Nx = 1001, \quad h = 0.001$

Use banded storage.

$Nx - 3.$

$$\therefore D = \begin{cases} X & X & x & -1 & -1 & -1 & \cdots & -1. \\ X & X & 4 & 4 & \cdots & & & 4 \\ X & -5 & -5 & \cdots & & & & -5 \\ 2 & 2 & \cdots & & & & & 2. \end{cases} \Rightarrow [X\,X\,2][X\,X\,-5\,2][X\,4\,-5\,2]\cdots[-1,4,-5,2]$$

Col1 Col2 Col3. ColN.

$[4 \times N_x]$ N_x.

```
S = p*{ -1, 4, -5, 2};
for (int i=0; i<N; i++)
    for (int j=0; j<ldh; j++)
        H[ i*ldh + j ] = S[j]
```

$i=0.\ j=0\ \ H[0] = X.$
$j=1\ \ H[1] =$
$j=2.\ \ H[2] =$
$j=3\ \ H[3] = 2$
$i=1.\ \ j=0\ \ H[1\times 4+0] = H[4].$

$k = 2$ Superdiagonals, Col, upper.

upper
$$\begin{cases} X & X & -1 & -1 & -1 & \cdots & -1 \\ X & 16 & 16 & 16 & \cdots & & 16 \\ -30 & -30 & \cdots & & & & -30 \end{cases}$$

lower.
$$\begin{cases} 16 & 16 & \cdots & & & 16 & X \\ -1 & -1 & \cdots & & -1 & X & X \end{cases}$$

```
double S[ldh] = { -1.0*p, 16.0*p, -30}
```

$S \Rightarrow \begin{pmatrix} X & X & -30 \end{pmatrix}, \begin{pmatrix} X & 16 & -30 \end{pmatrix}, \begin{pmatrix} 1, & 16, & -30 \end{pmatrix} \cdots \begin{pmatrix} -1, & 16, & -30 \end{pmatrix}$

$2+1=3.$

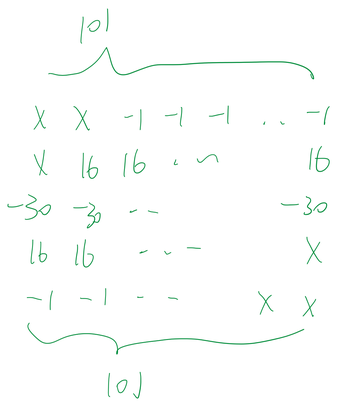
$D[i*ldh+j] = S[j].$

$$\begin{cases} D[0+0] = D(0) = X \\ D[0+1] = D[1] = 16 \\ D[0+2] = D[2] = -30. \end{cases}$$

$$\begin{cases} D[3+0] = D[3] = X \\ D[3+1] = D[4] = 16 \\ D[3+2] = D[5] = -30 \quad \checkmark \end{cases}$$

$D[6+0] = D[6] = S[0] = -1$
$D[6+1] = D[7] = S[1] = 16.$

$$p = \frac{1}{12h^2} = \frac{1}{(12)(0.01)(0.01)}$$



$\Rightarrow$

